CONTEXTE

PREP LIB RNASEQ (ANCHOR OU XT-HS2)

(CHRISTIAN DAVIAUD)

ILLUMINA STRANDED MRNA AND TOTAL RNA WORKFLOWS

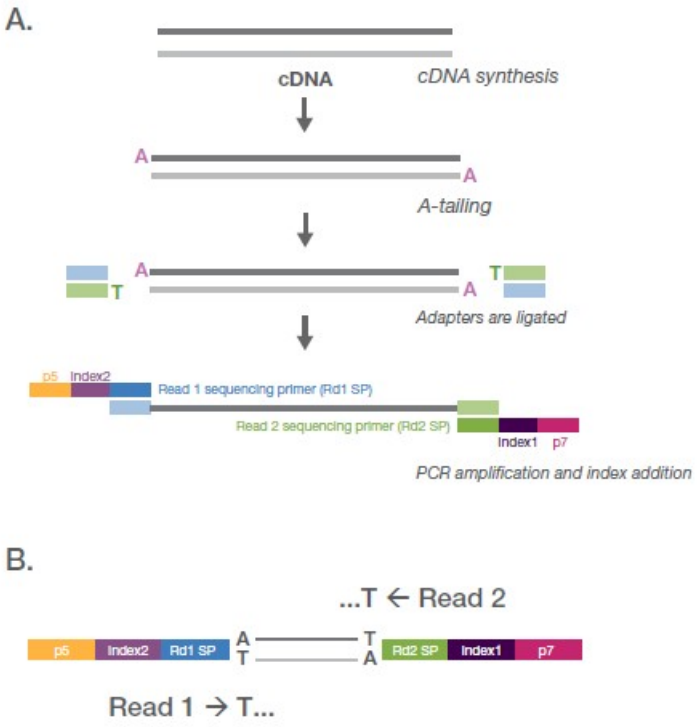


Figure 1: Illumina Stranded RNA library preparation—(A) After cDNA synthesis is complete, ligation of adapters and PCR amplification produces high-quality libraries. (B) The use of T-overhangs in sequencing adapters to facilitate rapid ligation results in all reads starting with a "T" in the first cycle.

- **Illumina Stranded mRNA Prep, Ligation** provides a cost-efficient option for coding RNA-focused analyses.
- **Illumina Stranded Total RNA Prep, Ligation with Ribo-Zero™ Plus** enables whole-transcriptome analysis, capturing coding and multiple forms of noncoding RNA.
- **Illumina RNA Prep with Enrichment** brings bead-linked transposome (BLT) technology to RNA enrichment.

Prepare Library | Sequence | **Analyze Data**

illumina

Best practices for read trimming for Illumina Stranded mRNA and Total RNA workflows

Explore the impact of the T-overhang on sequence read quality and options for read trimming.

Prepare Library | Sequence | **Analyze Data**

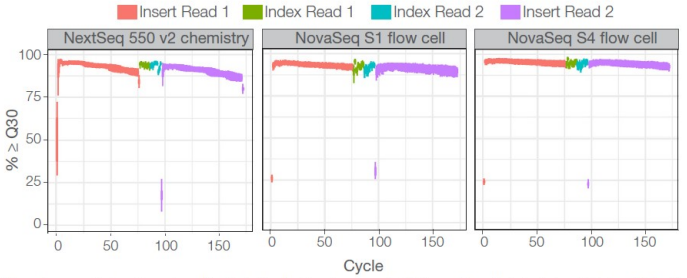


Figure 2: Impact of T-overhang on first cycle read quality—Including the T-overhang resulted in low read quality in the first cycle for Insert Read 1 and Insert Read 2 on the NextSeq 550 and NovaSeq 6000 Systems.

illumina-stranded-rna-t-overhang-tech-note-470-2020-010.pdf

Processus : Prep. Lib. pour séquençage Illumina > Prep Lib RNAseq (Anchor ou XT-HS2)



Prep. Librairie non indexée

Protocole
- Illumina Anchor strand totRNAseq RiboZ+
- Illumina Anchor strand mRNAseq
- SOP Agilent XT-HS2 mRNA lib Prep

Propriétés processus :
Type processus banque
- RK - ssmRNASeq-IllAnchor
- RL - sstRNASeq-IllAnchorRiboZ+
- RNA - XT-HS2 mRNA-Seq (trimming 2x5bases)



PCR + indexing

Protocole
- Illumina Anchor strand totRNAseq RiboZ+
- Illumina Anchor strand mRNAseq
- SOP Agilent XT-HS2 mRNA lib Prep

Plaques index (UDP0001-UDP0384)
IDT for Illumina Anchor DNA Unique Dual Indexes Set A
IDT for Illumina Anchor DNA Unique Dual Indexes Set B
IDT for Illumina Anchor DNA Unique Dual Indexes Set C
IDT for Illumina Anchor DNA Unique Dual Indexes Set D

Agilent SureSelect XT HS2 Kit A (index 1-96 plaque orange)
Agilent SureSelect XT HS2 Kit B (index 97-192 plaque bleue)
Agilent SureSelect XT HS2 Kit C (index 193-288 plaque verte)
Agilent SureSelect XT HS2 Kit D (index 289-384 plaque rouge)

QC LabChip GX

Lib normalisée

QC Quant-iT

QC Calcul Théor

Pool (satellite)

QC LabChip GX

Ou calcul moyenne de taille

QC QuBit

QC Calcul Théor

Normalisation (satellite)

Prep. FC ordonnée



Dépôt sur séquenceur

Process miniaturisé – volumes 2-3 µL